Vol 10 Chapter 9 — Special Functions

Keeper (author pass — deep math/physics revision)

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Chapter 9 — Special Functions

Level 1 — one sentence

Special functions — Gamma $\Gamma(s)$, zeta $\zeta(s)$, Bessel J_ν, Hermite H_n, Legendre P_ℓ, hypergeometric ${}_pF_q$, Fox H, Heckman-Opdam, mock theta — recur throughout physics and number theory, and the BST team's May 2026 special-functions investigation identifies substrate-natural function structure for the $\zeta_B(s)$ functional equation and Bergman-kernel hypergeometric forms.

Level 2 — graduate-physicist precision

9.1 Gamma function

$$\Gamma(s) = \int_0^\infty t^{s-1} e^{-t}\, dt \quad (\Re s > 0)$$

Extends factorial: $\Gamma(n) = (n-1)!$. Functional equation $\Gamma(s+1) = s\Gamma(s)$. Analytic continuation gives poles at $s = 0, -1, -2, \ldots$.

Stirling's formula: $\Gamma(s+1) \sim \sqrt{2\pi s}(s/e)^s$ as $s \to \infty$.

9.2 Riemann zeta function

$$\zeta(s) = \sum_{n=1}^\infty n^{-s} \quad (\Re s > 1)$$

Analytic continuation to $\mathbb{C} \setminus \{1\}$. Riemann Hypothesis (Clay Millennium): all non-trivial zeros on critical line $\Re s = 1/2$.

BST RH status: CONDITIONAL (April 21, 2026 three-leg proof; substrate ζ_B(s) functional equation proved T1638).

9.3 Hypergeometric functions

${}_2F_1(a, b; c; z) = \sum_n \frac{(a)_n (b)_n}{(c)_n n!} z^n$. Generalized: ${}_pF_q$.

Many physically-relevant special functions are special cases of hypergeometric. The Fox H function generalizes further.

9.4 Orthogonal polynomial families

Sturm-Liouville eigenfunctions (Ch 3) on various intervals with various weights: - Legendre $P_\ell(x)$ on $[-1,1]$: spherical harmonics - Hermite $H_n(x)$ on $\mathbb{R}$ with weight e^{-x^2}: harmonic oscillator - Laguerre $L_n^\alpha(x)$ on $[0,\infty)$ with weight $x^\alpha e^{-x}$: hydrogen radial - Chebyshev $T_n(x)$ on $[-1,1]$ with weight $1/\sqrt{1-x^2}$: minimax approximation - Jacobi $P_n^{(\alpha,\beta)}$: general two-parameter

9.5 Bessel and modified Bessel

$J_\nu(x), Y_\nu(x)$ (Bessel); $I_\nu(x), K_\nu(x)$ (modified). Appear in cylindrical symmetry.

9.6 BST special-functions investigation (May 2026)

The team's May 2026 special-functions investigation (memory `project_special_functions_investigation.md`) sought substrate-natural special function structure: - Fox H: partial results - Aleph functions: new candidate - Heckman-Opdam (multivariate Bessel): new candidate - Mock theta: new candidate - Bergman kernel on D_{IV}^5: explicit hypergeometric-type representation

Goal: recognize substrate's $\zeta_B(s)$ as a known special function → functional equation follows; further BST predictions become tractable.

9.7 K-audit anchors

- Vol 6 Ch 5 / Paper #9: heat-kernel arithmetic structure
- T1638: ζ_B functional equation proof
- May 2026 special-functions investigation (memory)

Level 3 — 5th-grader accessibility

Special functions: named functions appearing all over physics and number theory. Gamma $\Gamma(s)$: factorial extended. Zeta $\zeta(s) = 1 + 1/2^s + 1/3^s + \ldots$: Riemann's function (RH = where its zeros are). Hypergeometric ${}_2F_1$: master function many others are special cases of. Orthogonal polynomials (Legendre, Hermite, Laguerre, ...): eigenfunctions of physics problems. BST May 2026 investigation: looking for substrate-natural special functions for the substrate $\zeta_B(s)$.

What comes next

Chapter 10 develops calculus of variations + path integrals.

Where to look this up

- Whittaker and Watson, *A Course of Modern Analysis*
- Andrews-Askey-Roy, *Special Functions*
- BST: T1638, Paper #9, May 2026 SF investigation